

Supplementary Material for *The Fidelity–Utility Paradox in Surrogate-Based Reinforcement Learning for HVAC Control*

Supplementary Material Scope

This Supplementary Material provides notation, implementation formulas, supporting diagnostics, per-seed summaries, and per-regime tables for the results reported in the main article. It is intended as a reviewer-facing companion document. Numerical values needed to interpret the claims are reported directly in the main article and in the tables and figures below. The full project codebase is not publicly released at this stage. The building emulators are provided by the open-source BOPTEST framework (Blum et al., 2021; <https://github.com/ibpsa/project1-boptest>) and are deployed as a Docker container as described in the BOPTEST RTE section below.

Notation (read first). Throughout this Supplementary Material the legacy implementation labels **v3** and **v3.5** denote **BB** (the black-box surrogate) and **GB** (the grey-box surrogate) of the main article, respectively. Figure and table labels keep the legacy **v3/v3.5** names for consistency with the experimental records; all main-text prose uses **BB/GB**.

Supplementary Nomenclature

Symbol	Unit	Meaning
<i>Thermophysical quantities</i>		
T_{zone}	$^{\circ}\text{C}$	zone air temperature
T_{amb}	$^{\circ}\text{C}$	ambient air temperature
T_{sup}	$^{\circ}\text{C}$	supply-air temperature command
C_{zon}	J K^{-1}	zone thermal capacitance
R	K W^{-1}	lumped envelope thermal resistance
η	–	heating power conversion efficiency
$\dot{Q}$	W	net zone heat flow
P	W	total HVAC electrical power
Δt	s	control / integration step (900 s)
<i>Reinforcement-learning formulation</i>		
s_t, a_t, o_t, d_t	–	state, action, observation, disturbance at step t
Φ	–	one-step plant (emulator) transition map
f_{θ}, g_{ϕ}	–	black-box surrogate (v3) and frozen physical twin (v3.5)
r_t	–	per-step (shaped) reward
$\tilde{r}_t$	–	hybrid reward with disagreement censor
$J(\pi)$	–	expected discounted return
γ	–	discount factor
w_c, w_e, w_s	–	comfort / energy / safety preference weights
$\hat{A}_t$	–	generalized advantage estimate

Symbol	Unit	Meaning
ρ_t	–	PPO probability ratio
ϵ	–	PPO clipping parameter
δ	–	surrogate–twin disagreement penalty
λ_T, λ_P	–	hybrid temperature / power censor weights
<i>Evaluation metrics</i>		
m_s	–	operational maintenance/safety score evaluated on BOPTEST (lower is better)
$r_{\text{time}}, r_{\text{sev}}$	–	time-in-violation and severity components of m_s
$m_{s,\text{RL}}, m_{s,\text{PI}}$	–	operational score of the RL and built-in PI controller
RMSE_T	°C	live closed-loop zone-temperature RMSE
$\text{CV}(\text{RMSE})$	%	coefficient of variation of the RMSE
NMBE	%	normalized mean bias error
R^2	–	coefficient of determination
Violation	%	fraction of steps outside the comfort band
CV	–	coefficient of variation (std/mean) over seeds
g_a	–	L_2 action-gap norm (surrogate vs live)
Δm_s	–	live-minus-surrogate m_s transfer gap
<i>Transferability (testcase k)</i>		
τ	–	pre-specified pass threshold $1.25 m_{s,\text{PI}}$
ΔE	%	energy change of the RL controller vs PI
G_{RMSE}	%	rollout-RMSE improvement after Stage A/B/C recalibration
ρ_C	–	re-identified C_{zon} ratio vs the <code>bestest_air</code> baseline
$\mathcal{A}_k$	–	pre-specified actuator adapter
<i>Abbreviations</i>		
BOPTEST	–	Building Optimization Testing Framework
RTE	–	(BOPTEST) Run-Time Environment
DRL / PPO	–	deep RL / proximal policy optimization
GAE	–	generalized advantage estimation
HDRL	–	hierarchical deep reinforcement learning
MORL	–	multi-objective reinforcement learning
POMDP	–	partially observable Markov decision process
RC / ODE	–	resistance–capacitance / ordinary differential equation
PI / RBC / MPC	–	proportional–integral / rule-based / model predictive control

This document is the Supplementary Material for the above manuscript. It provides supporting definitions, diagnostics, configuration summaries, per-seed tables, and per-regime transferability results referenced from the main text. Tables S1–S3 summarize the three evidence blocks.

Table S1: Results I reviewer-facing support map.

Content	What is reported in this Supplementary Material
Surrogate construction	Data-corpus sizes, model-role summary, v3 configuration, feature scaling, and physical constraints.
Grey-box calibration	Stage A telemetry alignment, Stage B excitation filtering, identified C_{zon} , and Stage A/B/C calibration metrics.
Predictive validation	Multi-horizon rollout fidelity, episode-level replicability, physical-consistency diagnostics, and matched-corpus attribution.
Runtime feasibility	BOPTEST and surrogate throughput comparisons under the common 900 s control protocol.

Table S2: Results II reviewer-facing support map.

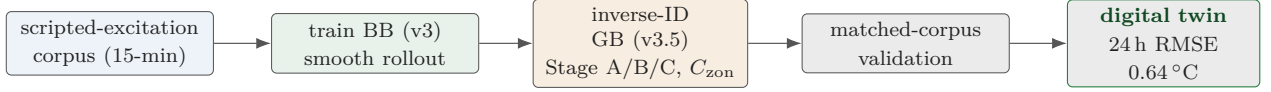
Content	What is reported in this Supplementary Material
Central controller comparison	Targeted-window KPIs for pure BB, direct GB, matched-resolution BB, and hybrid PPO.
Mechanism diagnostics	Action-surface smoothness, action saturation, transfer gaps, and closed-loop trace diagnostics.
Seed robustness	$N = 3$ thermostatic seed bands and collapse/usable verdict stability.
Controller-family boundary tests	HDRL censor-weight sweep, MORL observation-interface ablation, MORL Pareto summaries, and yearly PI comparison.

Table S3: Results III reviewer-facing support map.

Content	What is reported in this Supplementary Material
Transfer protocol	Target testcases, actuator adapters, pass thresholds, and recalibration regimes.
Controller-side transfer	Frozen-policy maintenance score, PI threshold comparison, and energy caveat by testcase.
Surrogate-side transfer	Raw versus full-recalibration rollout errors and C_{zon} re-identification across the hydronic family.
Hypothesis closure	Pre-specified predictions versus observed transfer outcomes.

Supplementary figures and tables

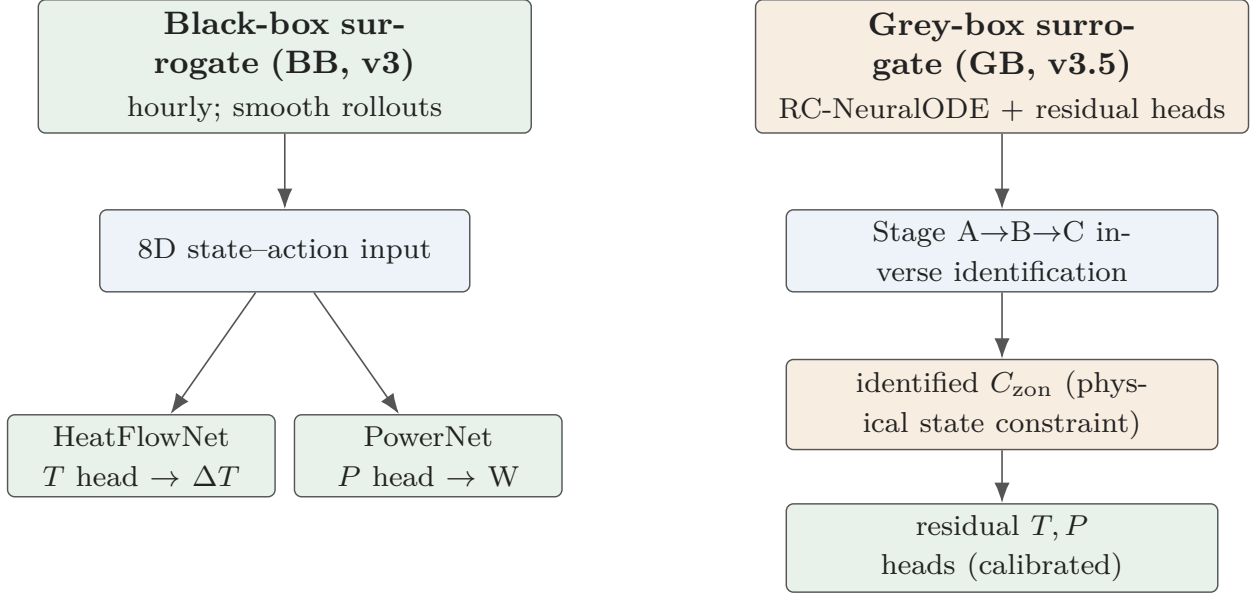
The following figures and tables provide supporting schematics, calibration and controller diagnostics, and detailed per-regime and per-seed breakdowns. They are cited from Results I–III (as Figure S/Table S references) but are not required to follow the main argument.



each step commits a versioned artefact (corpus, checkpoints, calibration summary, metrics) for reproducibility

Figure S1: Block 1 evidence chain. The figure reports the actual row counts, 24 h rollout errors, identified physical parameter, and speed benchmark values used to close Results I.

Schematic; not derived from a dataset.



BB supplies smooth rollout dynamics for policy training; GB is a physically identified twin used as a frozen reward censor.

Figure S2: Surrogate role separation inside Block 1. v3 is the compact dual-head rollout model; v3.5 adds Stage A/B/C physical identification and residual-head calibration.

Schematic; not derived from a dataset.

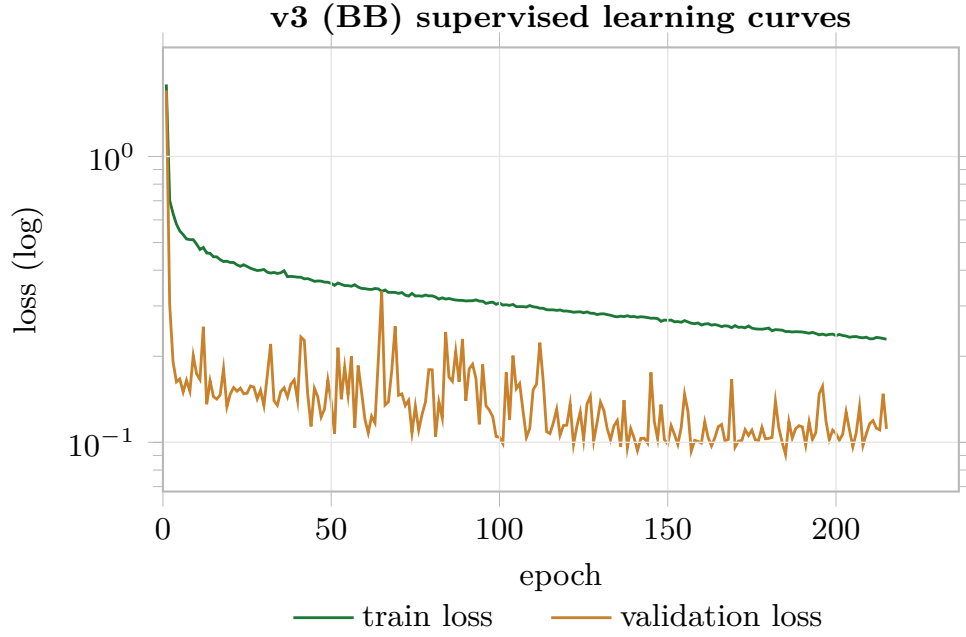


Figure S3: v3 supervised train/validation learning curves. Early stopping at the best validation epoch precedes any validation-loss upturn, indicating no overfitting at the selected checkpoint.

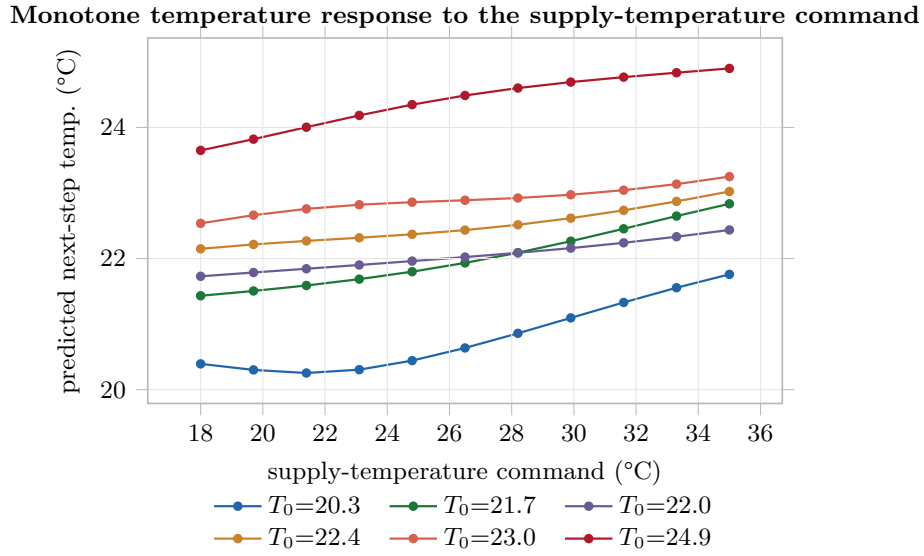


Figure S4: Physical-consistency audit: predicted next-step zone temperature versus the supply-temperature command for representative initial states. The response has the correct control sign and is predominantly increasing (mean Spearman 0.91), with small departures from strict monotonicity contributed by the neural residual head.

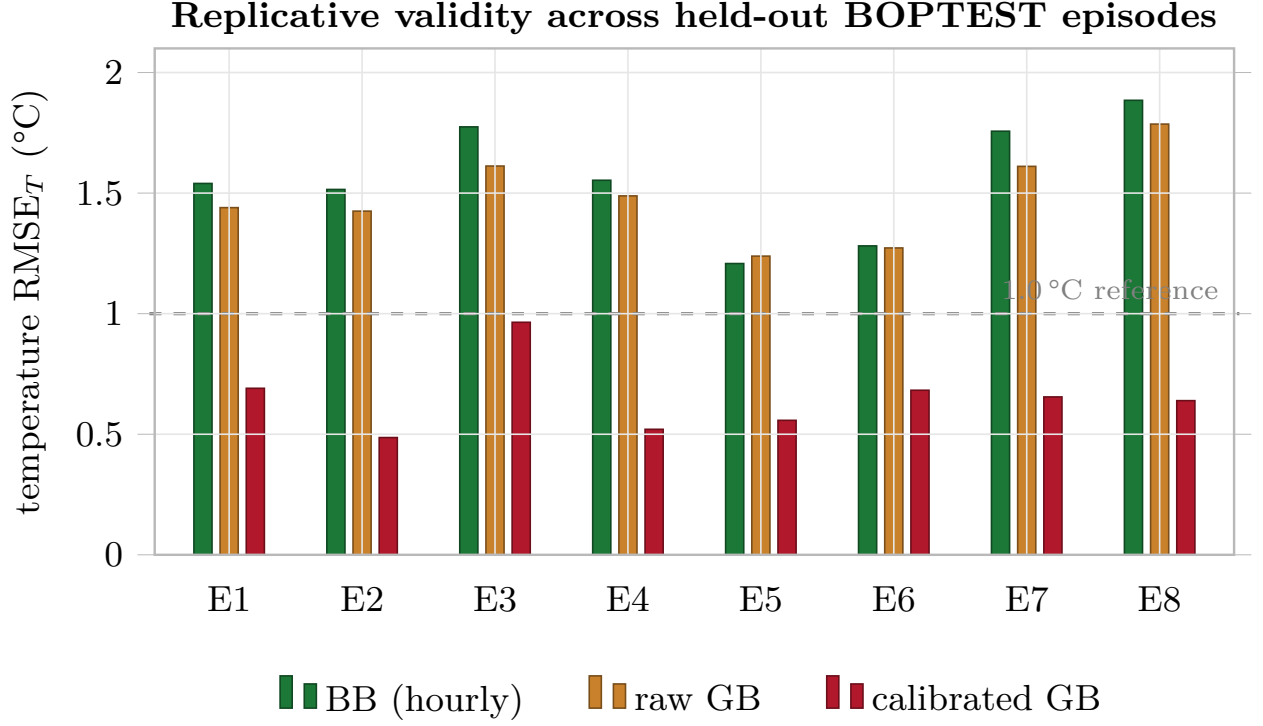


Figure S5: Replicative validity across the eight held-out BOPTEST episodes used by the Block 1 prepared-rollout results. The dashed line marks a 1 °C engineering reference.

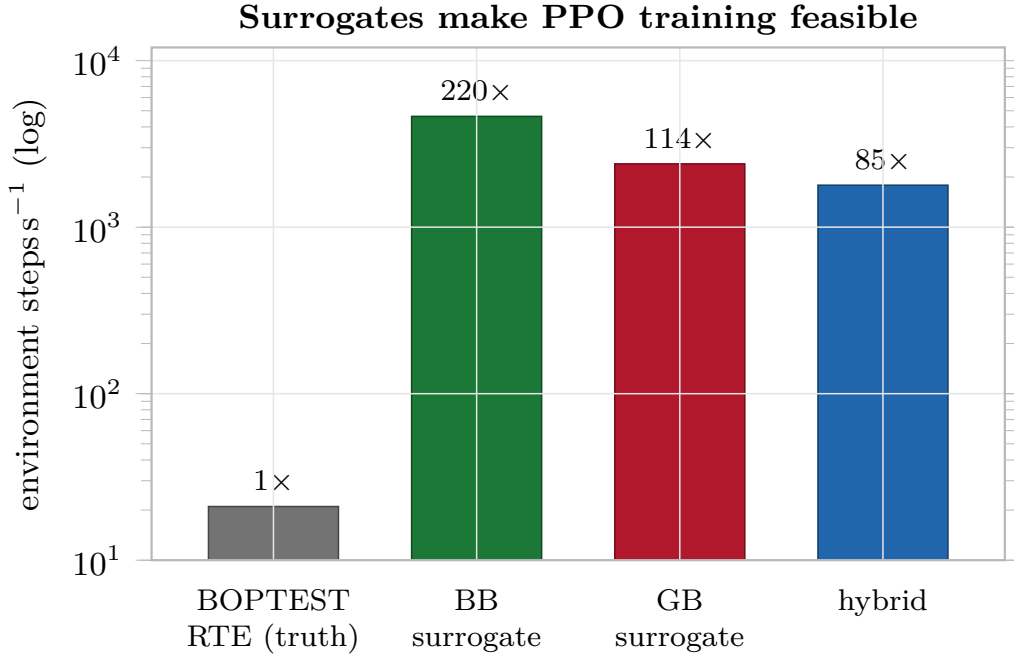


Figure S6: Runtime feasibility benchmark. The hybrid reference is included only to demonstrate downstream feasibility; controller utility is evaluated in Block 2.

Table S4: Block 1 data corpora referenced by the experimental protocol.

Dataset	Rows	Step (s)	Policy mix	Scenario mix
v3 canonical corpus	51,200	3600	cool, heat, mixed, random	autumn, spring, summer, winter
v3.5 calibration corpus	10,744	900	hdrl, thermostatic	peak_heat_window, typical_heat_window
15-min exploration corpus	48,384	900	cool, heat, mixed, pulse, random, thermostatic_noise	autumn, spring, summer, winter

Table S5: Architecture and calibration summary for the two Block 1 surrogate roles.

Surrogate	Role	Architecture / physics	Corpus	Training or calibration protocol
v3 direct-TSup	Fast control-oriented rollout dynamics	8D dual-head MLP: HeatFlowNetV2 (7,105 params, LayerNorm/Tanh/residual), PowerNetV2 (1,377 params, Tanh/Softplus); total 8,482 params	51,200 hourly transitions	AdamW, lr=1e-3, batch=256, epochs=500
v3.5 calibrated	Physics-informed predictive twin	RC-NeuralODE backbone with positive C_zon reparameterization and residual temperature/power heads	10,744 prepared 15-min transitions	Stage A latency/bias; Stage B 120 epochs; Stage C 60+80 epochs; C_zon=4.413e5 J/K

Table S6: v3 configuration from the active model class and Hou-and-Evins reporting results.

Item	Value
Input dimension	8
Temperature head	HeatFlowNetV2, 7,105 params
Power head	PowerNetV2, 1,377 params
Total parameters	8,482
Epoch budget	500
Batch size	256
Learning rate	1e-3
Optimizer	AdamW
Early stopping	patience=30; improvement_threshold=1e-6

Table S7: Feature scaling and physical constraints used by the Block 1 surrogate interface.

Variable	Scaling method	Parameters	Engineering role
surrogate_t_zone	affine min-max to [-1, 1]	min=15 C; max=35 C	Keeps the learned thermal state within the training support used by the direct-TSup surrogate.
surrogate_t_amb	affine min-max to [-1, 1]	min=-10 C; max=40 C	Normalizes outdoor-air temperature for the heat and power heads.
hour	cyclic sine/cosine	$\sin(2\pi \cdot \text{hour}/24)$; $\cos(2\pi \cdot \text{hour}/24)$	Avoids discontinuity between hour 23 and hour 0.
day	cyclic sine/cosine	$\sin(2\pi \cdot \text{day}/365)$; $\cos(2\pi \cdot \text{day}/365)$	Represents seasonal phase without an artificial year-end jump.
a0_raw	native action domain	[-1, 1] maps to 18..35 C supply temperature	Preserves the PPO action coordinate while keeping physical supply-temperature bounds explicit.
a1_raw	native action domain	[-1, 1] maps to 0..1 fan command	Keeps fan modulation aligned with the controller action range.
power_head_output	Softplus multiplied by P_MAX	P_MAX=5500 W	Guarantees non-negative predicted HVAC power.

Table S8: Stage A telemetry-alignment operations from the calibration-summary JSON.

Operation	Estimated value	Effect
Latency compensation	1 step (15 min)	latency-search RMSE 0.378 C
Temperature bias removal	-0.1049 C	median residual offset
Power affine normalization	scale -0.702, bias 1148 W	global power-channel rescale
Post-Stage-A temperature RMSE	0.2383 C	baseline for Stage B
Post-Stage-A power MAE	514.4 W	baseline for Stage C

Table S9: Stage B excitation filtering and physical identification of C_{zon} .

Parameter	Value	Notes
C_zon prior	4.200×10^5 J/K	air volume x specific heat
C_zon after Stage B	4.413×10^5 J/K	+5.1% vs prior
Stage B epochs	120	episode-aware run
C_zon learning rate	10^{-3}	dedicated scalar LR
Excitation quantile	0.95	dT threshold 0.1759
Excitation rows	403 / 8058	mean score 0.240 vs 0.069

Table S10: Stage A/B/C calibration summary from the canonical JSON and rollout results.

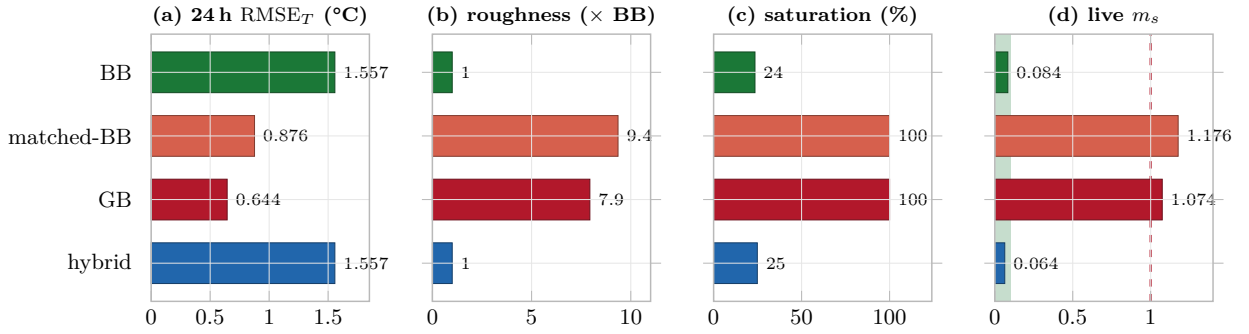
Metric	Before	After	Relative change
1-step RMSE_T ($^{\circ}\text{C}$)	0.384	0.235	-38.8%
24 h rollout RMSE_T ($^{\circ}\text{C}$)	1.466	0.644	-56.1%
Power MAE (W)	808	482	-40.3%
C_{zon} ($\times 10^5$ J/K)	4.200	4.413	+5.1%

Table S11: Multi-horizon teacher-forced rollout fidelity from the Hou-and-Evins predictive-validity table. Cells are temperature RMSE ($^{\circ}\text{C}$) at each horizon; the final column is the 24 h temperature R^2 (n/a where not computed for the auxiliary baselines).

Model	1 h	4 h	8 h	24 h	24 h R_T^2
v3 (hourly)	1.471	1.579	1.579	1.558	-1.411
v3 (15-min matched)	0.875	0.874	0.873	0.876	n/a
raw v3.5	1.491	1.491	1.491	1.466	n/a
calibrated v3.5	0.321	0.575	0.638	0.644	+0.588

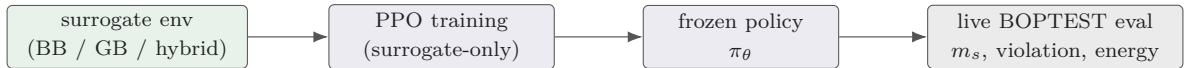
Table S12: Backend throughput under the 900 s control protocol from the speed-benchmark result (single CPU thread, BOPTEST reached through the same HTTP-Docker interface used in training).

Backend	Steps/s	Median (ms)	P95 (ms)	Speed-up
BOPTEST RTE HTTP loop	21.0	14.915	19.251	1.0 $\times$
v3 surrogate	4,626.3	0.165	0.273	220.2 $\times$
v3.5 calibrated	2,399.5	0.359	0.495	114.2 $\times$
hybrid v3+v3.5	1,786.8	0.507	0.648	85.0 $\times$



Quantified evidence chain: lower predictive RMSE_T does not imply lower live m_s

Figure S7: Quantified evidence chain for the fidelity-utility paradox. Each row is a training backend (v3 hourly, matched-resolution v3, direct v3.5, hybrid); columns report the measured links from predictive fidelity to live utility: **(A)** 24 h rollout RMSE_T ($^{\circ}\text{C}$), **(B)** scale-free action-surface roughness ($\times$ the usable v3), **(C)** policy saturation ($|a_0| \geq 0.9$, % steps), **(D)** live m_s with the $m_s = 1$ collapse line and usable band. The paradox is the contradiction between (A) and (D): the RMSE_T order ($\text{v3.5} < \text{matched-v3} < \text{v3}$) reverses in live m_s .



training never touches the live emulator; the policy is frozen before deployment (zero-shot to BOPTEST)

Figure S8: Block 2 controller-learning pipeline. The experiments separate the rollout model, the physical regularizer, the controller family, the observation interface, and live BOPTEST validation.

Schematic; not derived from a dataset.

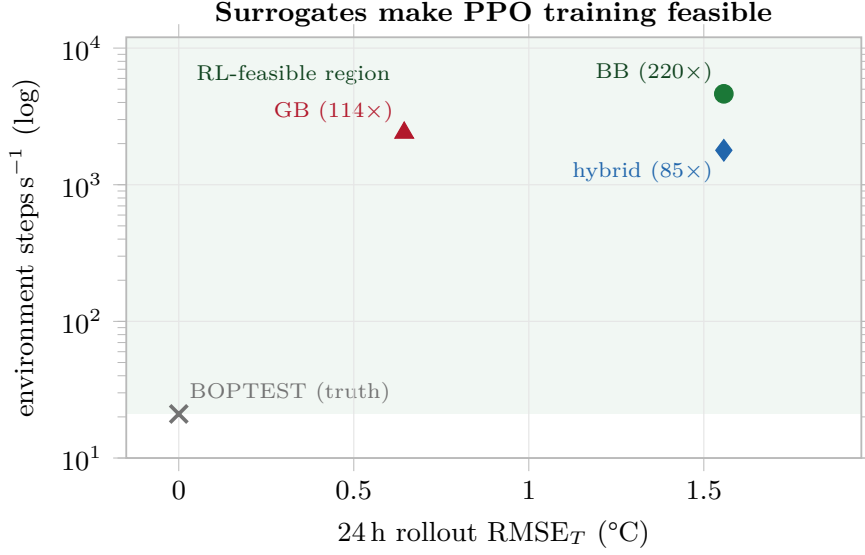
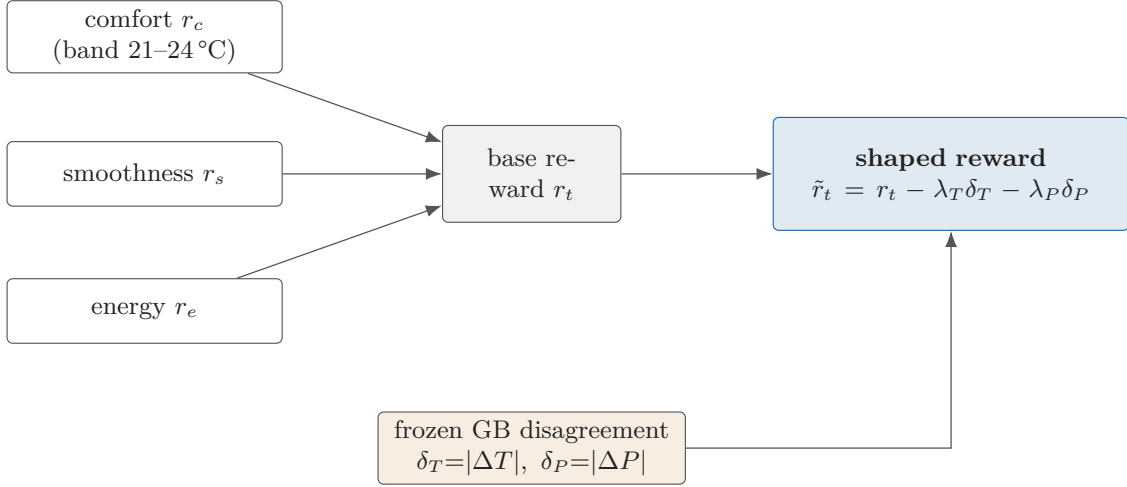


Figure S9: Surrogates make policy-gradient training computationally feasible. Environment throughput (env steps/s, log scale) versus 24 h rollout RMSE_T for the BOPTEST emulator (ground truth) and the three in-process surrogate backends. The surrogates run 85–220× faster than the BOPTEST RTE HTTP testbed, turning a days-long PPO run into hours; the hybrid is plotted at v3’s rollout RMSE because it rolls out on v3. The plotted values combine measured throughput with 24 h rollout-RMSE estimates.



$$\lambda_T = 0.10, \lambda_P = 5 \times 10^{-5}; \text{ the GB penalty is forward-only and never enters the policy loss.}$$

Figure S10: Hybrid reward-shaping mechanism. v3 supplies rollout dynamics; frozen v3.5 supplies a disagreement penalty on the same state-action transition.

Schematic; not derived from a dataset.

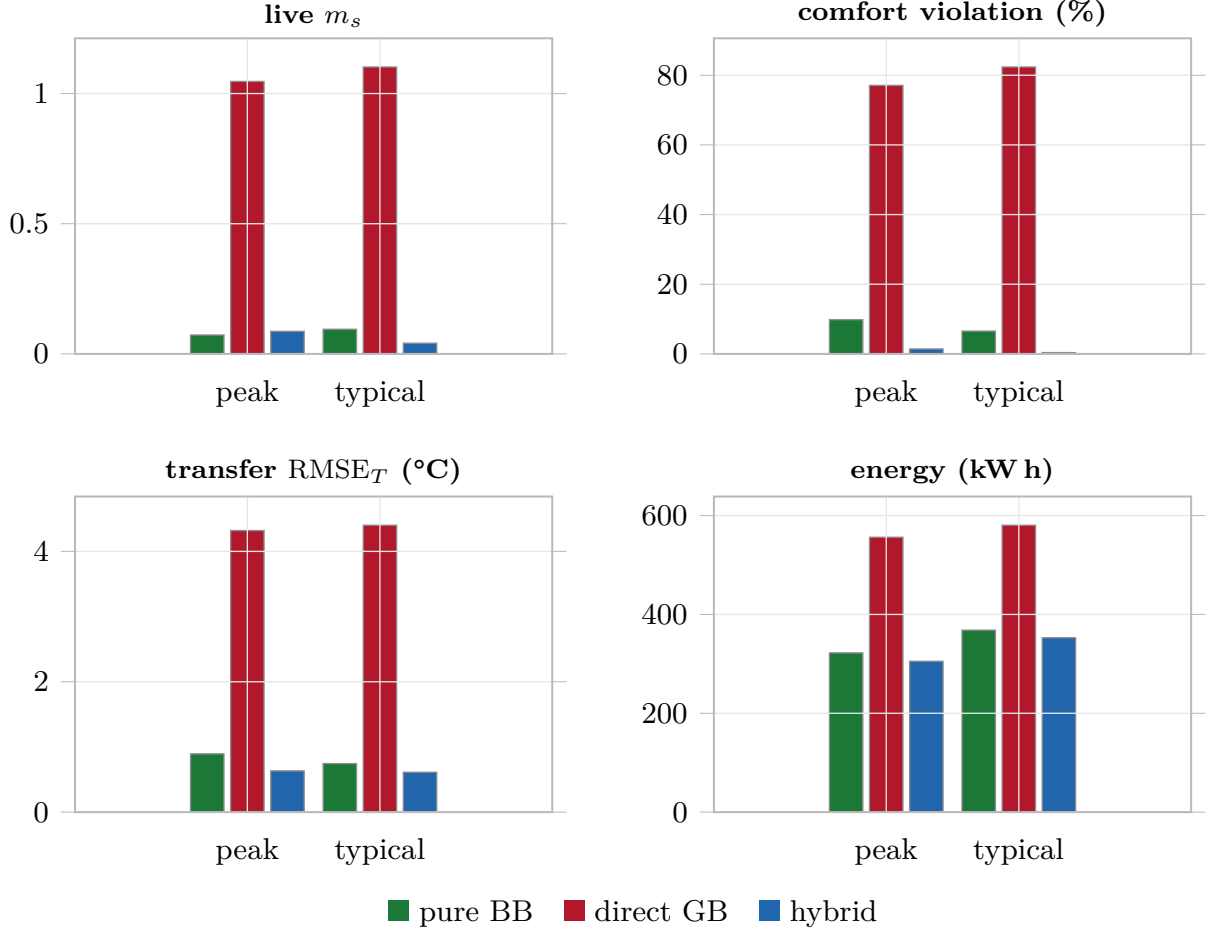


Figure S11: Detailed live BOPTEST KPI breakdown (per-metric bars) for pure v3, direct v3.5, and hybrid PPO. Direct v3.5 is a negative control: higher predictive fidelity does not imply a useful RL training environment. These per-metric bars give the underlying m_s , violation, RMSE, and energy behind the compact fidelity–utility scatter in the main text.

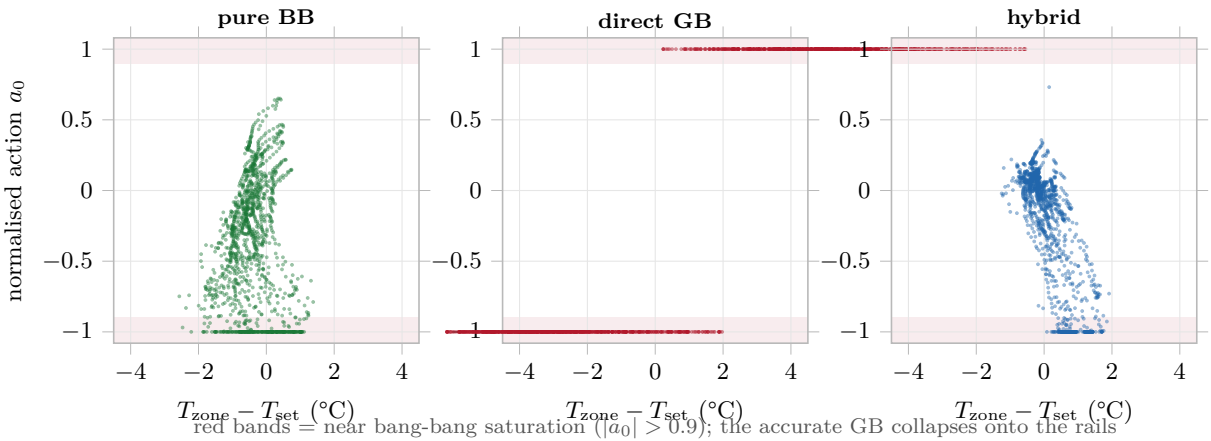


Figure S12: Action-density phase portrait of the three thermostatic controllers on the live BOPTEST `typical_heat_window`: empirical hexbin density of the normalised supply-temperature action a_0 (one hue per backend) against the thermal error $T_{\text{zone}} - T_{\text{set}}$, on *identical* axes, with the marginal a_0 distribution in the right strip of each panel and the measured saturation share ($|a_0| \geq 0.9$). The red bands mark the actuator-saturation region. Direct v3.5 collapses onto $a_0 = -1$ (100% saturation, a bang-bang law) whereas the usable hourly v3 and the hybrid modulate (≈ 19 – 20%).

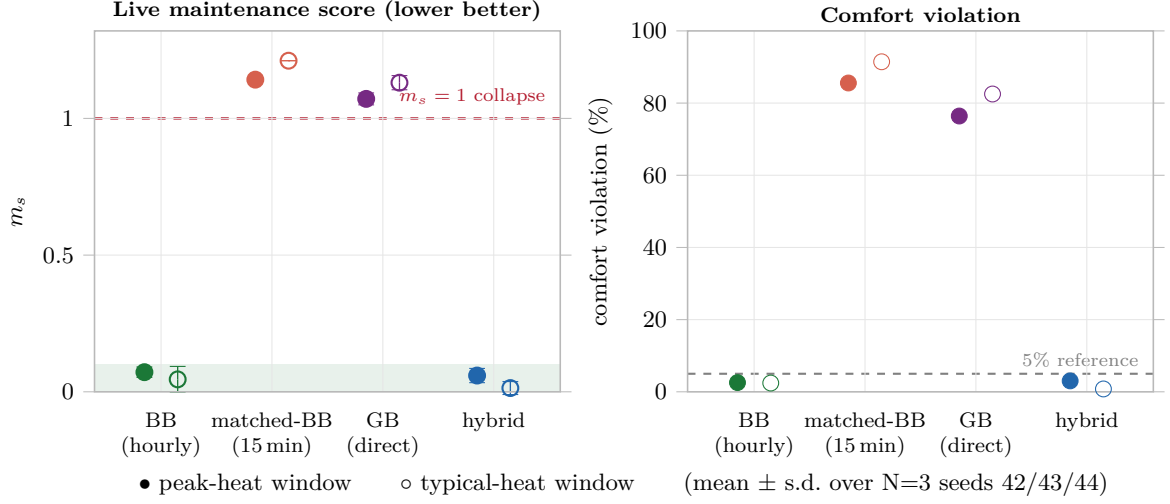


Figure S13: Across $N = 3$ seeds the fidelity–utility paradox is not a seed result: the usable hourly-v3 and hybrid controllers stay below both the $m_s = 1$ closed-loop-collapse line and the 5% comfort-violation reference on every seed, while the matched-resolution v3 and the direct v3.5 both collapse on every seed ($m_s > 1$, $> 75\%$ violation) with small seed variance. Markers are the mean over seeds {42,43,44} (filled = peak-heat window, open = typical-heat window); whiskers are $\pm$ one standard deviation; colour encodes the controller (green = coarse v3, salmon = matched-resolution v3, purple = direct v3.5, blue = hybrid). **(left)** live maintenance score m_s (lower is better, with the usable $m_s < 0.1$ band shaded); **(right)** comfort-violation fraction. Same data as Table S17.

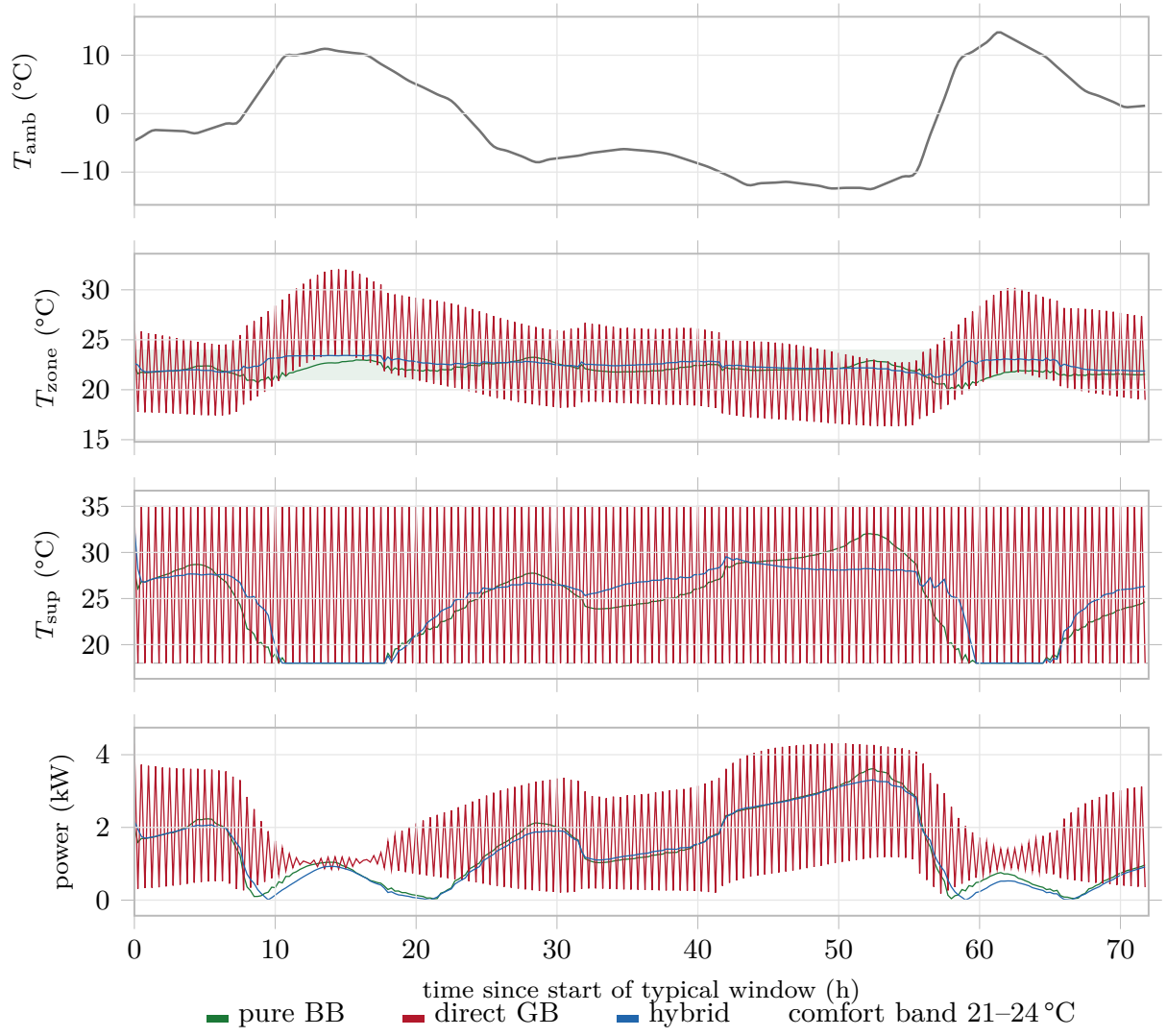


Figure S14: Closed-loop BOPTEST traces with ambient disturbance, comfort band, physical actuator limits, and HVAC power. This diagnostic visualizes the time-domain behaviour behind the compact main-paper roughness and maintenance-score figures.

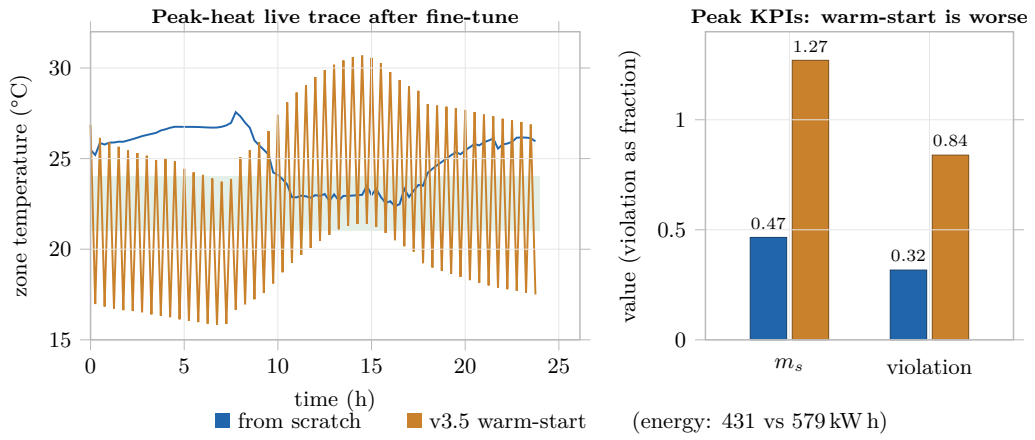
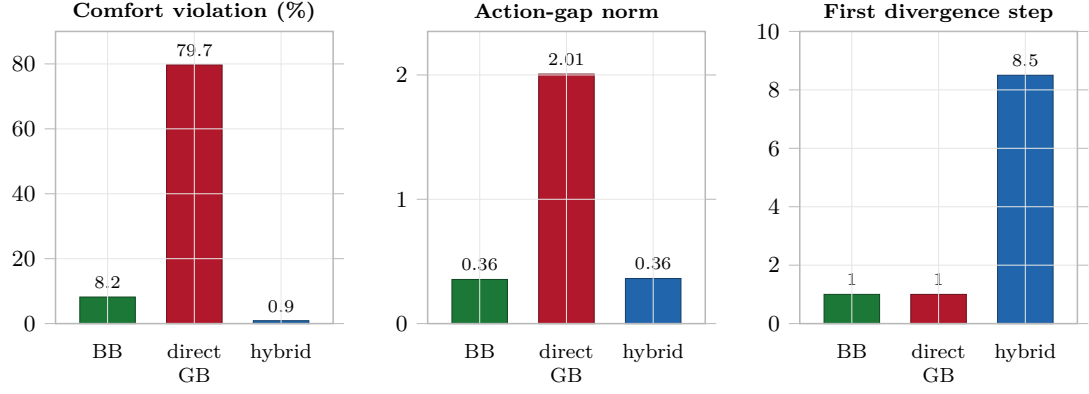
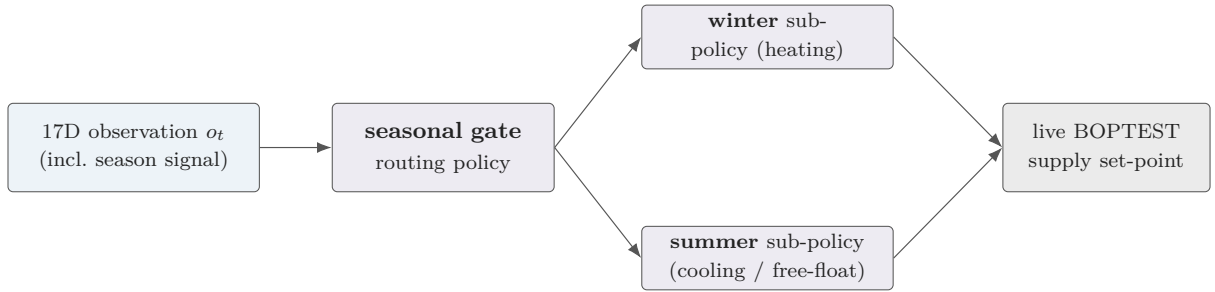


Figure S15: Warm-start negative control. Pretraining on direct v3.5 and then fine-tuning on the hybrid backend is inferior to training the hybrid policy from scratch.



Transfer-gap diagnostics: standalone direct-GB training saturates (bang-bang)

Figure S16: Transfer-gap diagnostics. Direct v3.5 has high action-gap norm and live-surrogate mismatch; the hybrid backend suppresses both.



The gate selects a season-specialised sub-policy; the disagreement penalty λ_T that helps thermostatic PPO does not transfer to this family (controller-family specific).

Figure S17: HDRL architecture: a high-level seasonal gate $k(t)$ routes the 17D observation to one of two low-level PPO setpoint specialists (winter / summer). Because the low level is already comfort-aware, the v3.5 temperature sensor over-regularizes it.

Schematic; not derived from a dataset.

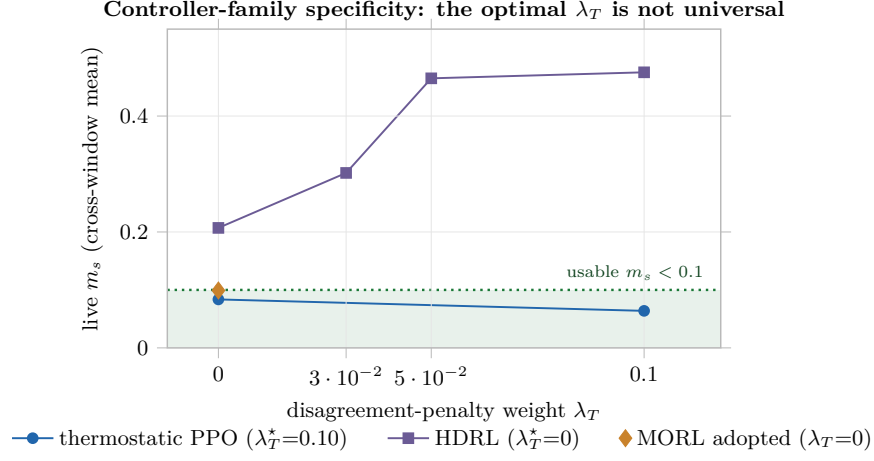


Figure S18: The optimal censor weight λ_T is controller-family specific. Live BOPTEST maintenance score m_s (cross-window mean of the peak and typical windows) versus λ_T for each controller family; $\star$ marks the per-family optimum. Thermostatic PPO improves with the censor (optimum $\lambda_T = 0.10$, the canonical hybrid), HDRL is best *without* it ($\lambda_T = 0.00$) and degrades monotonically, and MORL adopts $\lambda_T = 0.00$. HDRL has a dense four-point sweep; PPO is shown at its two measured endpoints (pure v3 = $\lambda_T 0$, hybrid = 0.10) and MORL at its single adopted configuration. The panel compares the *location* of each optimum, not absolute m_s across families (they use different observation/action interfaces).

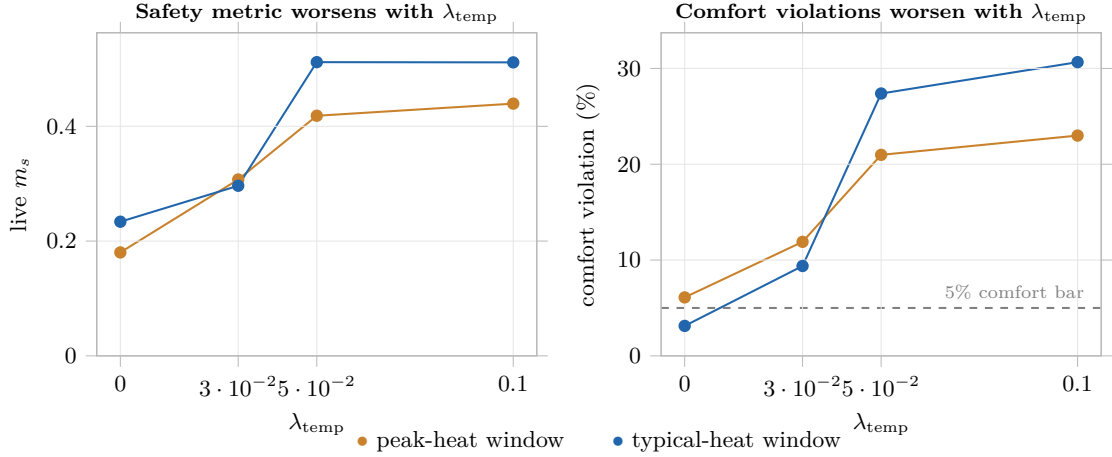
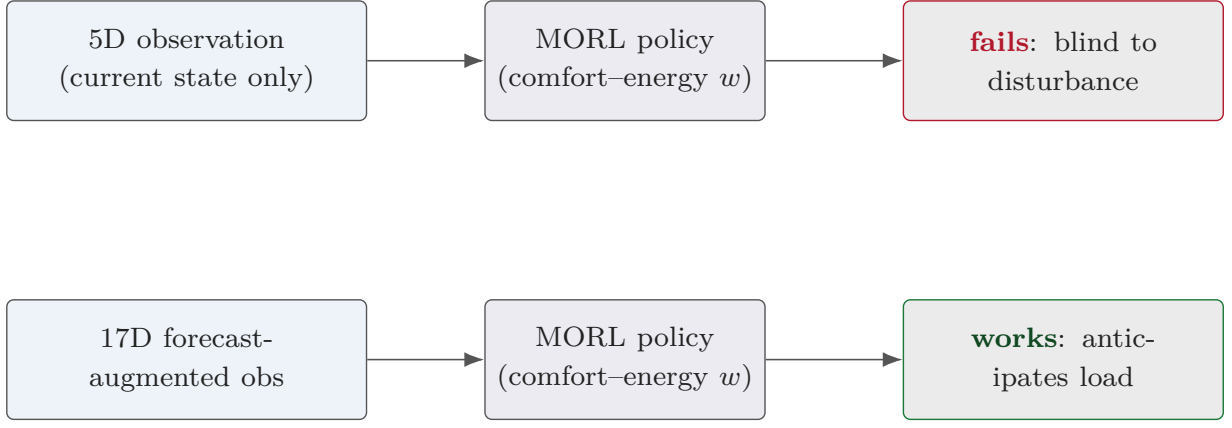


Figure S19: HDRL λ_T sensitivity. The thermostatic regularization weight does not transfer; the best HDRL policy uses no temperature-disagreement term.



The 5D→17D observation-interface change (adding forecast context), not the reward weighting, is what makes MORL control-viable.

Figure S20: MORL four-stage pipeline: surrogate pretrain → ERAM preference adaptation → live-BOPTEST finetune → yearly evaluation, all on the 17D power-only hybrid backend. The preference vector conditions the scalarized reward.

Schematic; not derived from a dataset.

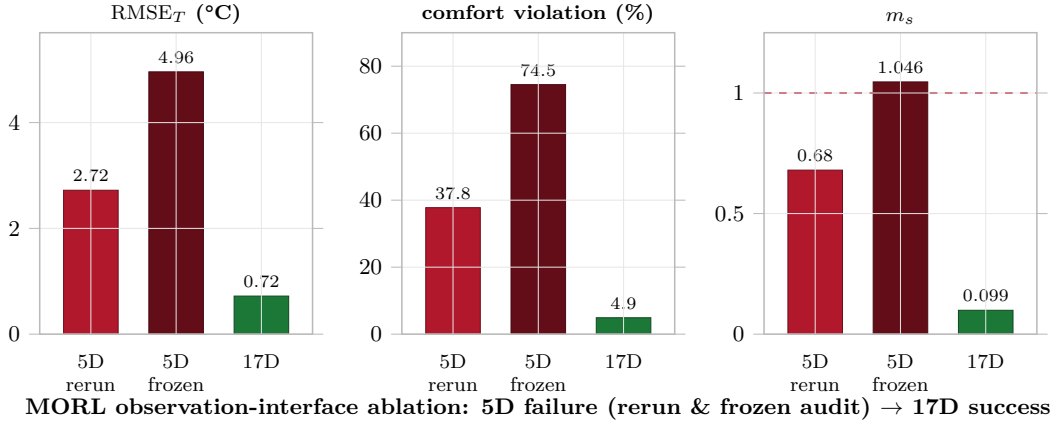


Figure S21: MORL observation-interface ablation. The main article reports the headline 5D-versus-17D numbers in its controller-family boundary-test paragraph; this supplementary plot includes the current-code 5D rerun, the frozen historical 5D audit result, and the 17D power-only recovery.

MORL 17D yearly seasonal validation

violation (%)	5.4	2.5	3.7	0.5	0	0.3	2.2	2.3	8.3	11.7	13.2	8.3
energy (kW h)	403	397	320	250	173	137	125	134	167	193	264	419
live m_s	0.14	0.06	0.08	0.02	0	0.02	0.08	0.08	0.16	0.2	0.2	0.14
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec

Figure S22: MORL seasonal performance heatmap under the 17D interface. The figure shows monthly structure but does not support the N=3 mechanism claim after N=5 extension.

MORL seasonal variance after N=5 falsification (across-seed m_s std)

Practical $w=(0.75, 0.25)$	0.124	0.188	0.106	0.171	0.091	0.075	0.076	0.089	0.023	0.058	0.121	0.181
Neutral $w=(0.50, 0.50)$	0.114	0.143	0.089	0.136	0.098	0.057	0.057	0.09	0.033	0.099	0.132	0.154
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec

Figure S23: Post-N=5 seasonal variance diagnostic. The earlier seasonal-inversion hypothesis is reported as falsified, not retained as a positive mechanism claim.**Table S13:** Targeted-window and yearly scenario definitions.

Scenario	Day idx	Start (s)	Dur. (d)	Ambient ($^{\circ}\text{C}$)	Role
peak_heat_window	3	259200	14	-24.4	January coldest, heating stress test
typical_heat_window	37	3196800	14	2.4	February moderate, deployment-realistic
yearly evaluation	12 months	—	14/mo	varied	MORL + PI yearly summary

Table S14: PPO training configuration used in Block 2.

Parameter	Thermostatic PPO	HDRL	MORL pretrain	MORL finetune
Algorithm	PPO, MlpPolicy	PPO, MlpPolicy	PPO, MlpPolicy	PPO load+continue
Learning rate	3.0×10^{-4}	3.0×10^{-4}	3.0×10^{-4}	1.0×10^{-4}
n_steps	1024	1024	2048	inherited
batch_size	4096	2048	64	inherited
n_epochs	10	10	10	10
γ	0.99	0.99	0.99	0.99
GAE λ	0.95	0.95	0.95	0.95
Clip range	0.20	0.20	0.20	0.20
Total timesteps	10M	12M	2M	100k
Seed	42	42	42	42–46 (N=5)

Table S15: Extended 17D observation feature groups.

Feature group	Dim.	Contents
Physical state (basic)	5	T_{zone} , CO_2 , clipped-log power, prev. T_{sup} , T_{amb}
Cyclic time	4	hour and day sine/cosine encodings
Ambient forecast	5	T_{amb} at +1, +3, +6, +12, +24 h
Previous action	2	$(a_{T_{\text{sup}}}, a_{\text{fan}})$ from last step
Temperature delta	1	causal-smoothed ΔT_{zone}
Total (extended)	17	obs_mode = extended

Table S16: Reward-shaping parameters.

Component	Value	Role
Comfort band	21.0 / 24.0 C	comfort interval (band_low/high)
Comfort deadband	0.50 C	soft margin around band edges
In-band bonus	0.050/step	reward for staying inside the band
Undershoot / overshoot weight	1.15 / 1.15	generic violation penalties
Cold-ambient asymmetry	1.60 (amb < 8.0 C)	extra cold-undershoot penalty
Hot-ambient asymmetry	1.80 (amb > 24.0 C)	extra hot-overshoot penalty
Heating action bonus	0.04 ($T_{\text{sup}} \geq 29.0$ C)	anti-degenerate shaping
Cooling action bonus	0.06 ($T_{\text{sup}} \leq 21.0$ C)	anti-degenerate shaping
MORL weights ($w_c/w_e/w_s$)	0.80 / 0.20 / 0.00	canonical scalarization
Energy scale	$2e - 4$ (W $\rightarrow$ reward)	energy-to-reward conversion

Table S17: Seed robustness ($N = 3$ seeds {42,43,44}) of the thermostatic controllers on the live BOPTEST targeted windows; mean $\pm$ standard deviation. The usable controllers (pure v3, hybrid) stay below the 5% comfort-violation bar on every seed, while the collapse controls (matched-resolution v3, direct v3.5, and the Δt -rescaled hourly v3) exceed $m_s = 1$ on every seed with small seed variance. The Δt -rescaled control keeps the hourly v3 but rescales its increment to the 900 s runtime step; its collapse shows the hourly BB’s usability rests on its large coarse-scale increment.

Controller	Window	N	m_s (mean $\pm$ std)	Violation % (mean $\pm$ std)
pure v3	peak	3	0.072 ± 0.019	2.6 ± 1.2
pure v3	typical	3	0.045 ± 0.047	2.4 ± 2.2
matched v3 (15-min)	peak	3	1.142 ± 0.000	85.6 ± 0.0
matched v3 (15-min)	typical	3	1.211 ± 0.000	91.4 ± 0.0
direct v3.5	peak	3	1.071 ± 0.022	76.4 ± 0.7
direct v3.5	typical	3	1.131 ± 0.026	82.5 ± 0.7
Δt -rescaled v3	peak	3	1.187 ± 0.010	83.2 ± 0.7
Δt -rescaled v3	typical	3	1.215 ± 0.006	86.0 ± 0.3
hybrid ($\lambda_T=0.10$)	peak	3	0.059 ± 0.026	3.0 ± 1.6
hybrid ($\lambda_T=0.10$)	typical	3	0.014 ± 0.024	0.8 ± 1.4

Table S18: Measured response-surface smoothness of the three single-model backends, probed directly on each surrogate’s one-step action map (no controller, no BOPTEST). The scale-free relative roughness (curvature normalised by slope) is independent of the step length and tracks closed-loop usability: the usable hourly v3 is the smoothest; both collapsed backends are several times rougher.

Surrogate backend	Slope $ \partial \hat{T} / \partial a_0 $	Rel. roughness	vs. v3	Closed-loop role
canonical v3 (hourly)	3.368	0.169	1.0 (ref)	usable
matched v3 (15-min)	0.272	1.578	9.4 $\times$	collapse
calibrated v3.5	0.467	1.338	7.9 $\times$	collapse

Table S19: Thermostatic hybrid λ_T sweep *design* (experimental protocol §5; $\lambda_P = 5 \times 10^{-5}$ throughout); the mid point is the retained canonical. Of these pre-registered brackets, the live-BOPTEST two-window evaluation reported in the paper covers thermostatic PPO at $\lambda_T = 0.10$ (canonical hybrid) and $\lambda_T = 0$ (pure BB), so the controller-family λ_T figure shows PPO at these two measured endpoints; the 0.05 and 0.15 brackets were not carried to full live two-window evaluation. The dense four-point λ_T sweep is the HDRL family, not PPO.

λ_T	Tag	Role
0.05	hybrid_l005	weaker censor (sweep bracket)
0.10	hybrid_l010	canonical (retained; $m_s = 0.087$ peak, 0.041 typical)
0.15	hybrid_l015	stronger censor (sweep bracket)

Table S20: Direct-v3.5 warm-start utility.

Mode	Scenario	m_s	Violation (%)
scratch (random init)	peak	0.465	31.70
scratch (random init)	typical	0.578	45.54
warm-start (from v3.5)	peak	1.270	83.93
warm-start (from v3.5)	typical	1.289	86.83

Table S21: Transfer diagnostics across three backends. $\Delta m_s = \text{surrogate} - \text{live}$ (negative $\Rightarrow$ surrogate optimistic); Temp gap is the live closed-loop RMSE_T .

Variant	Scenario	Temp gap ($^{\circ}\text{C}$)	Δm_s	Action gap	First div.	Top driver
pure v3	peak	0.894	-0.102	0.377	1	p_total_norm
pure v3	typical	0.745	-0.068	0.333	1	p_total_norm
hybrid $\lambda_T = 0.10$	peak	0.633	-0.023	0.473	1	p_total_norm
hybrid $\lambda_T = 0.10$	typical	0.612	-0.021	0.253	16	p_total_norm
direct v3.5	peak	4.320	-0.886	2.000	1	t_zone_norm
direct v3.5	typical	4.401	-1.014	2.014	1	t_zone_norm

Table S22: HDRL sweep over λ_T on the targeted windows; best is $\lambda_T = 0$, and performance degrades monotonically as the temperature censor strengthens.

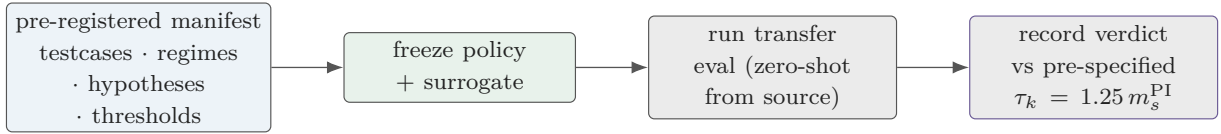
λ_T	Scenario	m_s	Violation (%)	RMSE_T ($^{\circ}\text{C}$)	Energy (kWh)
0.00	peak	0.180	6.10	0.751	329.6
0.03	peak	0.307	11.90	0.993	311.5
0.05	peak	0.418	20.98	1.303	298.1
0.10	peak	0.440	22.99	1.343	300.4
0.00	typical	0.234	3.12	0.691	385.1
0.03	typical	0.296	9.38	0.959	369.6
0.05	typical	0.512	27.38	1.491	354.5
0.10	typical	0.511	30.65	1.455	357.1

Table S23: MORL Pareto (seed 42) and N=5 canonical seed analysis.

Preference / statistic	RMSE _T (°C)	Violation (%)	m_s	Interpretation
0/100 (seed 42)	8.359	86.76	1.588	energy-only collapse
25/75 (seed 42)	0.912	11.39	0.173	energy-weighted usable
50/50 (N=5 mean±std)	0.893±0.081	13.01±6.62	0.187±0.078	neutral, CV=0.42
75/25 (N=5 mean±std)	0.799±0.100	9.23±6.67	0.139±0.085	practical, CV=0.61
100/0 (seed 42)	0.631	1.49	0.032	best seed-42 comfort

Table S24: Per-seed MORL yearly metrics for the two canonical weight pairs.

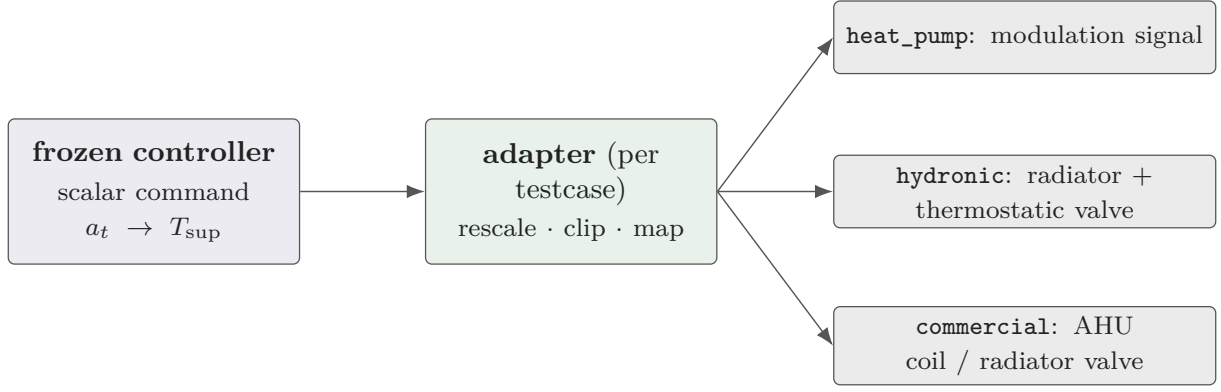
Pair (c/e)	Seed	RMSE _T	Within 1°C (%)	Violation (%)	Energy (kWh)	m_s
50/50	42	0.939	74.9	12.92	2843.6	0.193
50/50	43	0.769	82.9	6.77	2872.0	0.103
50/50	44	0.879	76.9	9.34	2858.4	0.141
50/50	45	0.895	77.7	11.97	2839.8	0.187
50/50	46	0.985	65.4	24.05	2554.1	0.310
75/25	42	0.700	83.3	3.08	2994.3	0.057
75/25	43	0.716	84.9	5.47	2963.9	0.087
75/25	44	0.769	81.7	7.82	2819.1	0.117
75/25	45	0.903	78.5	9.42	3036.7	0.160
75/25	46	0.909	70.3	20.37	2602.8	0.276



the manifest is git-tagged before any evaluation (audit anchor) — no post-hoc threshold tuning

Figure S24: Block 3 pre-specified transferability protocol. Controller-side transfer is tested against a testcase-specific PI threshold; surrogate-side transfer is tested by full Stage A/B/C recalibration on target telemetry.

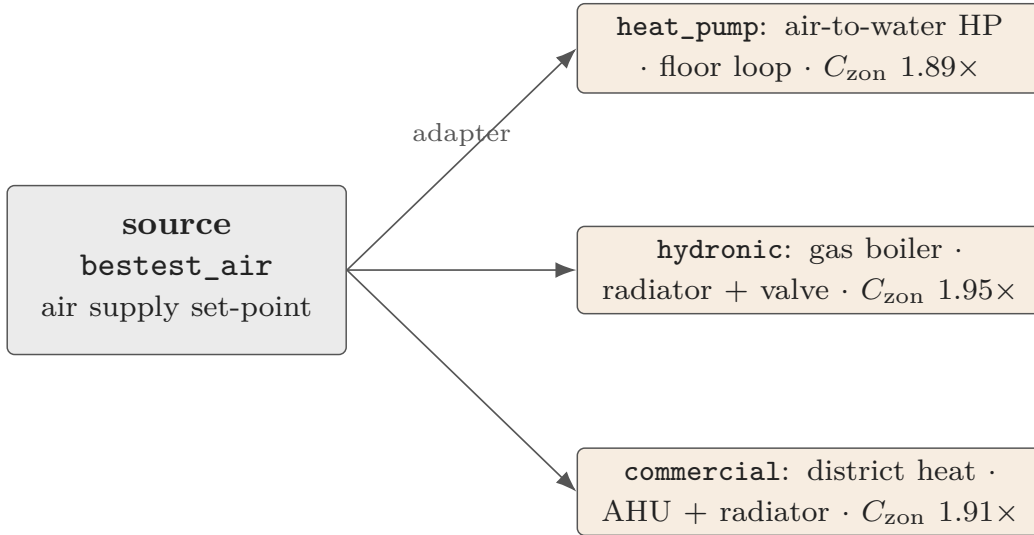
Schematic; not derived from a dataset.



Transfer is adapter-mediated: the frozen policy's scalar set-point is re-mapped to each test case's actuator interface — not a literal copy of the source actions.

Figure S25: Adapter-mediated transfer schematic. The frozen Block 2 hybrid policy is routed through a pre-specified actuator adapter $\mathcal{A}_k$ to each testcase's actuator interface; the controller-side verdict (against the $1.25\times$ PI threshold) is per-testcase, while full Stage A/B/C surrogate recalibration passes on all three.

Schematic; not derived from a dataset.



the hydronic family shares a near-constant C_{zon} ratio ($\approx 1.9\times$), a family regularity exploited for transfer.

Figure S26: Schematic (no data). Comparative HVAC topology of the source case and the three hydronic targets (source $\rightarrow$ distribution $\rightarrow$ zone). Transfer changes the heat source, distribution path, and actuator set, while the single-zone energy balance and the zone thermal capacitance C_{zon} are shared across the family.

Schematic; not derived from a dataset.

Frozen-controller transfer verdict and m_s^{RL}/τ

	none	partial	full
hydronic heat pump	FAIL $m_s/\tau = 1.15$	FAIL $m_s/\tau = 1.15$	FAIL $m_s/\tau = 1.15$
hydronic	FAIL $m_s/\tau = 1.04$	FAIL $m_s/\tau = 1.04$	FAIL $m_s/\tau = 1.04$
commercial	COND PASS $m_s/\tau = 0.55$	COND PASS $m_s/\tau = 0.55$	COND PASS $m_s/\tau = 0.55$

identical across surrogate-recalibration regimes: the controller is frozen

Figure S27: Controller-transfer verdict heatmap. Residential hydronic cases fail the controller threshold; the commercial case passes the threshold but carries a separate energy caveat.

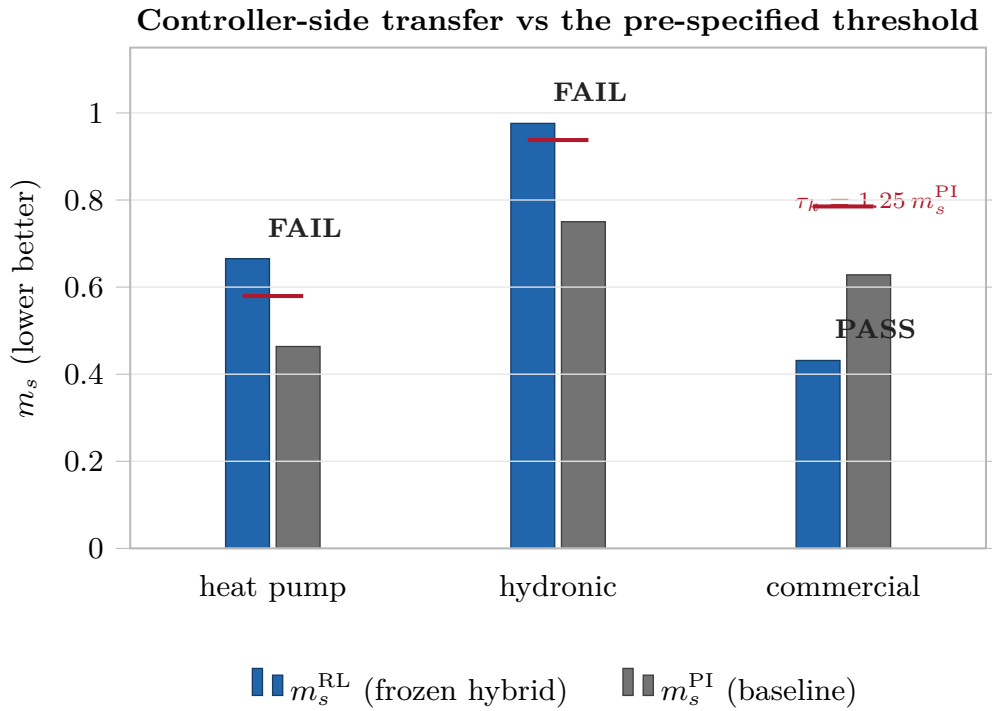
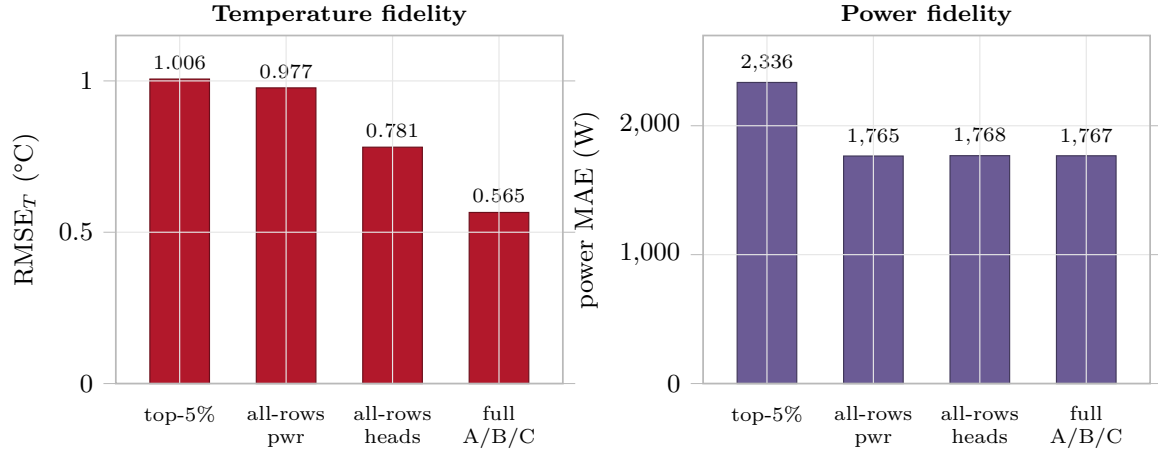


Figure S28: Controller-side transfer: frozen-hybrid $m_{s,\text{RL}}$ against the PI baseline $m_{s,\text{PI}}$ and the pre-specified threshold $\tau = 1.25 m_{s,\text{PI}}$ (dashed). The two residential cases exceed the threshold (FAIL); the commercial case is below it (PASS, with the separate energy caveat).



Surrogate fidelity improves with recalibration depth (controller frozen)

Figure S29: Primary-testcase surrogate-fidelity progression across recalibration regimes. Temperature and power fidelity improve monotonically toward full Stage A/B/C, while the frozen controller’s live m_s is invariant by experimental design.

Hypothesis closure vs version-locked predictions

Hypothesis / prediction	Observed closure	Evidence
● H1 strong	FALSIFIED	frozen recipe, direct transfer
● H2 medium	FALSIFIED (by structural definition)	partial recalibration
● H3 surrogate	SUPPORTED (on N=3)	full Stage A/B/C RMSE
● H3 controller	FALSIFIED (regime-dependent failure)	controller full regime
● Stretch none	expected FAIL → observed PASS	commercial, mode=none
● Stretch C_{zon}	expected scale-dep. → observed 1.909×	commercial C_{zon}
● Stretch RMSE	expected 50–90% → observed 87.81%	commercial recalibration

Figure S30: Hypothesis closure matrix: Block 3 supports surrogate-side transferability but falsifies deployment-ready frozen-controller transfer.

Table S25: Pre-specified target testcases and actuator adapters.

Role	Testcase	Structural difference vs <code>bestest_air</code>	Adapter
Primary	<code>bestest_hydronic_heat_pump</code>	hydronic loop driven by a heat pump	<code>hydronic_t_supply_v1</code>
Secondary	<code>bestest_hydronic</code>	boiler/radiator hydronic distribution	<code>hydronic_direct_supply_v1</code>
Stretch	<code>singlezone_commercial_hydronic</code>	order-of-magnitude larger commercial zone + hydronic valve	<code>commercial_hydronic_valve_v1</code>

Table S26: Per-testcase actuator-adapter mapping. All adapters share $h = \text{clip}((T_{\text{sup},t} - 18)/17, 0, 1)$; setpoint overrides are in kelvin.

Testcase	Primary override	Override formula	Auxiliary overrides
heat_pump	oveTSet_u (K)	$288.15 + 9h$ (zone setpoint 15–24°C)	oveHeaPumY_u, ovePum_u, oveFan_u from h
hydronic	oveTSetSup_u (K)	$T_{\text{sup},t} + 273.15$ (direct supply)	ovePum_u from h ; fixed comfort-band context
commercial	dh_oveTSetSupSetHea_u (K)	$T_{\text{sup},t} + 273.15$ (district supply)	heating valves + pump from h ; fixed zone/air context

Table S27: Pre-specified recalibration regimes. The controller is frozen in every regime.

Regime	Surrogate action	Controller action
none	frozen bestest_air surrogate recipe	frozen Block 2 hybrid controller
partial	Stage C residual-head recalibration only	frozen Block 2 hybrid controller
full	Stage A + Stage B + Stage C on target telemetry	frozen Block 2 hybrid controller

Table S28: Testcase architecture and identified physics. C_{zon} in 10^5 J/K; equivalent mass = $C_{\text{zon}}/c_{p,\text{air}}$; mean delivered power from the Stage-C telemetry.

Testcase	Heat source	C_{zon}	eq. mass (kg)	mean power (W)	Thermal regime
bestest_air (source)	air supply	4.413	439	—	reference
hydronic heat pump	heat pump	8.348	831	2,778	conduction-dominated
hydronic (boiler)	boiler/radiator	8.623	858	679	slow hydronic
commercial hydronic	district + AHU	8.426	838	76,860	ventilation/AHU-dominated

Table S29: Primary-testcase per-regime detail. m_s is invariant across rows because the controller is frozen by experimental design; only surrogate fidelity changes.

Regime / result	RMSE _T (°C)	Power MAE (W)	$m_{s,\text{RL}}$	Status
PI baseline (yearly)	—	—	0.464	reference
none: frozen controller	—	—	0.665	control FAIL
partial: Stage C top-5%	1.006	2336	0.665	diagnostic
partial: all-rows power head	0.977	1765	0.665	diagnostic (power)
partial: all-rows heads	0.781	1768	0.665	surrogate cond. pass
full: Stage A/B/C	0.565	1767	0.665	surrogate PASS, ctrl FAIL

Table S30: C_{zon} re-identification across the **bestest_air** baseline and the three hydronic testcases.

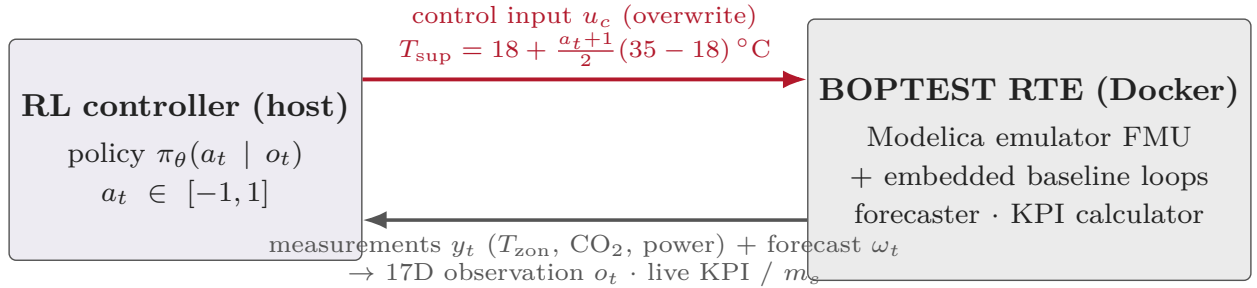
Testcase	C_{zon} (J/K)	ρ_C vs bestest_air
bestest_air (Block 1)	4.413×10^5	1.000 (baseline)
hydronic heat pump	8.348×10^5	1.892
hydronic (boiler)	8.623×10^5	1.954
commercial hydronic	8.426×10^5	1.909
hydronic-family mean $\pm$ std	—	1.918 ± 0.032

Table S31: Stretch-testcase pre-specified predictions versus observed outcomes.

Prediction	Pre-reg.	A-priori	Observed	Verdict
mode=none controller verdict	FAIL	0.80	threshold PASS	FALSIFIED
full surrogate RMSE gain	[50, 90]%	0.70	87.8%	CONFIRMED
C_{zon} hyp. A (uniform)	[1.7, 2.2] $\times$	0.35	1.91 $\times$	CONFIRMED
C_{zon} hyp. B (scale-dep.)	[3, 10] $\times$	0.50	1.91 $\times$	FALSIFIED
C_{zon} hyp. C (failure)	non-convergence	0.15	converged (N=3)	FALSIFIED

BOPTEST RTE control loop and supply-temperature actuation

The live evaluation runtime is the BOPTEST Run-Time Environment (RTE): a Dockerised Modelica emulator with embedded baseline local-loop and supervisory controllers, a disturbance forecaster, and a KPI calculator, all exposed through a RESTful HTTP API (Blum et al., 2021). The test controller does not replace the plant; following the BOPTEST formalism it *overwrites* a chosen subset u_c of the emulator’s control inputs while the remaining embedded loops keep running at baseline. In this study u_c is the single supply-temperature set-point of each test case, activated through the BOPTEST overwrite pair (the point’s `_activate` flag set to 1 and its `_u` value written); on `bestest_air` this point is the supply-air-temperature set-point `oveTSetSup`. At every control step $\Delta t = 900$ s (Option-1 synchronisation, in which the controller defines the step and the emulator advances when an action is returned) the controller reads measurements y_t (zone temperature, CO₂, power) and disturbance forecasts ω_t from the API, maps its policy output $a_t \in [-1, 1]$ to the supply-temperature command $T_{\text{sup}} = 18 + \frac{a_t+1}{2}(35 - 18)^\circ\text{C}$, writes it to the overwrite signal, and issues `POST /advance` to integrate the emulator forward one step. The RTE then returns the next measurements and the live key performance indicators (including the maintenance score m_s) computed from the closed-loop trajectory rather than from the surrogate. For the hydronic transfer cases the same supply-temperature command is routed through the per-test-case adapter to that case’s overwrite signal (heat-pump modulation, valve position, or supply-water set-point), so the frozen policy interface is unchanged across test cases (Figure S31).



one control step $\Delta t = 900$ s (Option-1 synchronisation via `POST /advance`)

Figure S31: BOPTEST RTE closed-loop control interface (topology after Blum et al., 2021, Figs. 1–3). At each 900 s step the policy overwrites only the supply-temperature set-point through the `oveTSetSup` signal pair and advances the Dockerised emulator, which returns measurements, forecasts, and the live maintenance score m_s .

Detailed formulas

The following standard or implementation-level equations were moved here to keep the main article focused on results; they are reproduced for completeness. **PPO clipped surrogate objective**

(Schulman et al., 2017), with $\hat{A}_t$ the GAE advantage:

$$L_{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right], \quad \rho_t(\theta) = \frac{\pi_\theta(a_t | o_t)}{\pi_{\theta_{\text{old}}}(a_t | o_t)}. \quad (1)$$

Black-box (BB) multi-horizon training loss with physical-consistency penalty:

$$\mathcal{L}_{\text{BB}} = \sum_{k \in \{1, 2, 4\}} \frac{1}{N} \sum_i (\hat{T}_i(t+k) - T_i(t+k))^2 + \lambda_{\text{phys}} \text{ReLU}(|\Delta \hat{T}| - \Delta T_{\text{max}}). \quad (2)$$

Grey-box (GB) positive reparameterization of the zone capacitance:

$$C_{\text{zon}} = c_s (\text{softplus}(\rho) + c_{\text{min}}/c_s), \quad C_{\text{zon}} \geq c_{\text{min}} = 5 \times 10^4 \text{ J/K}. \quad (3)$$

Stage B maximum-a-posteriori identification of C_{zon} on the excitation subset $\mathcal{E}$ under a weak Gaussian prior:

$$\hat{C}_{\text{zon}} = \arg \min_C \sum_{i \in \mathcal{E}} (\hat{T}_i(C) - T_i)^2 + \frac{1}{2\sigma_p^2} (C - C_{\text{prior}})^2. \quad (4)$$

Warm-started GB heat-term scaling: when a GB surrogate is warm-started from legacy BB weights, the inherited heat term is rescaled so that the per-step increment is dimensionally consistent at the runtime control step (the BB update lacks this scaling because its increment carries no explicit Δt):

$$s = s_{\text{legacy}} \frac{C_{\text{init}}}{\Delta t_{\text{legacy}} q_{\text{scale}}}. \quad (5)$$

Transfer-gap diagnostics: surrogate–live maintenance-score mismatch and mean action-gap norm:

$$\Delta m_s = m_{s, \text{surrogate}} - m_{s, \text{BOPTEST}},$$

$$g_a = \frac{1}{N} \sum_{t=1}^N \|a_{t, \text{BOPTEST}} - a_{t, \text{surrogate}}\|_2. \quad (6)$$

Tables relocated from the main article

The following tables were moved here to reduce the float load of the main article; they are cited from it as Supplementary Tables.

Table S32: Maintenance-score decomposition across controller families (cited in the main article’s controller-family synthesis). Thermostatic/HDRL are 14-day windows; MORL/PI are yearly means.

Controller (window/eval)	m_s	r_{time}	r_{sev}
pure BB (peak)	0.073	0.015	0.058
direct GB (peak)	1.046	0.771	0.276
hybrid $\lambda_T=0.10$ (peak)	0.087	0.047	0.040
HDRL $\lambda_T=0$ (peak)	0.180	0.061	0.119
pure BB (typical)	0.095	0.044	0.051
direct GB (typical)	1.102	0.824	0.278
hybrid $\lambda_T=0.10$ (typical)	0.041	0.024	0.017
HDRL $\lambda_T=0$ (typical)	0.234	0.031	0.202
MORL 50/50 (yearly, N=5)	0.187	0.130	0.057
MORL 75/25 (yearly, N=5)	0.139	0.092	0.047
PI (yearly)	0.910	0.636	0.274

Table S33: Controller-learning falsifiable-hypothesis ledger (CL1–CL5), cited in the main article’s central fidelity–utility test. These are distinct from the paper-level hypotheses H1–H4: CL1–CL3 are the controller-experiment counterparts of H1–H3, and CL4–CL5 add MORL-specific checks. Each hypothesis is tested on the live BOPTEST RTE.

	Hypothesis	Evidence	Verdict
CL1	The higher-fidelity twin (GB) used directly as the rollout environment yields the best controller.	central test	FALSIFIED (direct GB $m_s > 1.0$)
CL2	Calibrated GB is useful when its role changes from dynamics provider to a frozen reward-shaping censor.	hybrid remedy	SUPPORTED (hybrid gives best cross-window robustness)
CL3	The thermostatic-optimal censor weight $\lambda_T = 0.10$ transfers to the HDRL family.	HDRL sweep	FALSIFIED (HDRL best at $\lambda_T = 0$)
CL4	MORL viability is determined by the reward scalarization alone.	MORL 5D/17D	FALSIFIED (needs the 17D observation)
CL5	The single-seed MORL canonical ($m_s \approx 0.10$) is representative.	MORL N=5	FALSIFIED at N=5 (best of five; CV 0.42–0.61)

Table S34: Modelling assumptions of the direct supply-temperature (T_{sup}) control interface (cited in the main article’s predictive-validation section).

Assumption	Statement
Actuator abstraction	Supply-air temperature is commanded directly; valve/coil dynamics are not modelled beyond the Stage A latency term.
Command bounds	$T_{\text{sup}} \in [18, 35]^\circ\text{C}$; fan command $u_{\text{fan}} \in [0, 1]$.
Power non-negativity	$\hat{P} = \text{Softplus}(\cdot) \geq 0$, capped at $P_{\text{max}} = 5500$ W.
State support	Zone and ambient temperatures normalized to $[-1, 1]$ over $[15, 35]$ and $[-10, 40]^\circ\text{C}$.
Interface invariance	All four backends share the same transition signature; only the internal dynamics differ.